DS-A3: Regression Baseline and Diagnostics

Experimental Temperature Prediction and Plagioclase–Hornblende Geothermometer Errors

Research question

How reliably do the PH94 and PH21 geothermometers reconstruct experimental temperatures, and how do their errors vary across studies and with pressure, temperature, and mineral chemistry?

Study scope

This report contains two related but distinct analyses: a regression experiment predicting experimental temperature, and a comparison of PH94 and PH21 thermometer errors. Experimental temperatures are the reference values for assessing thermometer estimates.

Data

The audited S2 dataset contains 90 experimental runs and 60 columns. The thermometer comparison uses 65 matched runs. The regression analysis uses 65 complete-case observations and 19 predictors.

Introduction

Mineral-based geothermometers are used to estimate temperatures from mineral compositions, but their estimates may differ from experimentally measured temperatures. Quantifying these differences is important for understanding the reliability and limitations of geothermometric methods.

This study evaluates the PH94 and PH21 plagioclase–hornblende geothermometers against experimental temperatures and examines whether their errors vary across published studies and experimental conditions. In a complementary regression analysis, Linear Regression, Ridge, and Lasso are compared with a naive baseline to assess how well experimental temperatures can be predicted from pressure and mineral chemistry.

Why this analysis matters

A geothermometer's accuracy cannot be assessed only by examining a few estimated

temperatures. Its errors must be compared against experimental reference temperatures across different conditions and published studies. Positive errors indicate overestimation, whereas negative errors indicate underestimation. Mean absolute error (MAE) measures the magnitude of errors, while root mean squared error (RMSE) gives greater weight to large errors.

Data science approach

The regression component provides a complementary assessment of how experimental temperature relates to pressure and measured mineral chemistry. A naive mean-temperature baseline establishes a reference level of predictive performance. Linear Regression, Ridge, and Lasso are then compared to examine whether a more complex model provides useful predictive gains. Preprocessing and model selection must be performed without using the held-out test observations.

Diagnostics and interpretation

Predictive metrics alone cannot reveal every weakness of a model. Residual plots, influence diagnostics, learning curves, heteroscedasticity testing, and error comparisons across experimental conditions help identify where predictions are less reliable. The study also examines whether thermometer errors are associated with pressure, temperature, mineral chemistry, and experimental source.

Scope and limitations

The analysis is based on a relatively small dataset of experimental runs drawn from multiple studies. Observations from the same study may share experimental characteristics, so study-group separation is important when evaluating predictive performance. Associations between mineral chemistry and prediction errors are exploratory and do not establish causal relationships. Findings from experimental data should not automatically be generalised to natural geological samples.

1. Methods

During model selection, the naive baseline predicts the mean experimental temperature calculated from the training partition only. For final held-out test evaluation, the Ridge model and naive baseline both use the development data, comprising the training and validation partitions. The test partition is excluded from model fitting, baseline calculation, and model selection. This explains why the baseline prediction differs between validation and final test evaluation.

Model selection was based on validation and training cross-validation rather than test-set performance. The final held-out test set was evaluated after model selection. Regression diagnostics included residual inspection, influence, learning curves, and a Breusch–Pagan test.

Thermometer errors are calculated relative to experimental temperature. Signed error indicates direction of bias; mean absolute error (MAE) measures average error magnitude. Exploratory correlations are treated as associations rather than causal effects.

Split and leakage audit

split	study_id	n_runs
train	alonso-perez_2009	5
train	blatter_2001	1
train	caricchi_2006	5
train	gardner_1995	2
train	grove_2003	1
train	pichavant_2002	6
train	pietranik_2009	3
train	prouteau_scaillet_2003	7
train	scaillet_1999	7
train	schmidt_1996	1

Validation model comparison

model	MAE_C	RMSE_C	R2	best_alpha	best_CV_MAE_C
Ridge	43.806	55.476	0.531	1.000	29.553
Lasso	47.374	54.262	0.551	0.010	31.170
Linear Regression	47.415	54.225	0.551		
Naive baseline	61.217	89.516	-0.222		

2. Regression Diagnostics

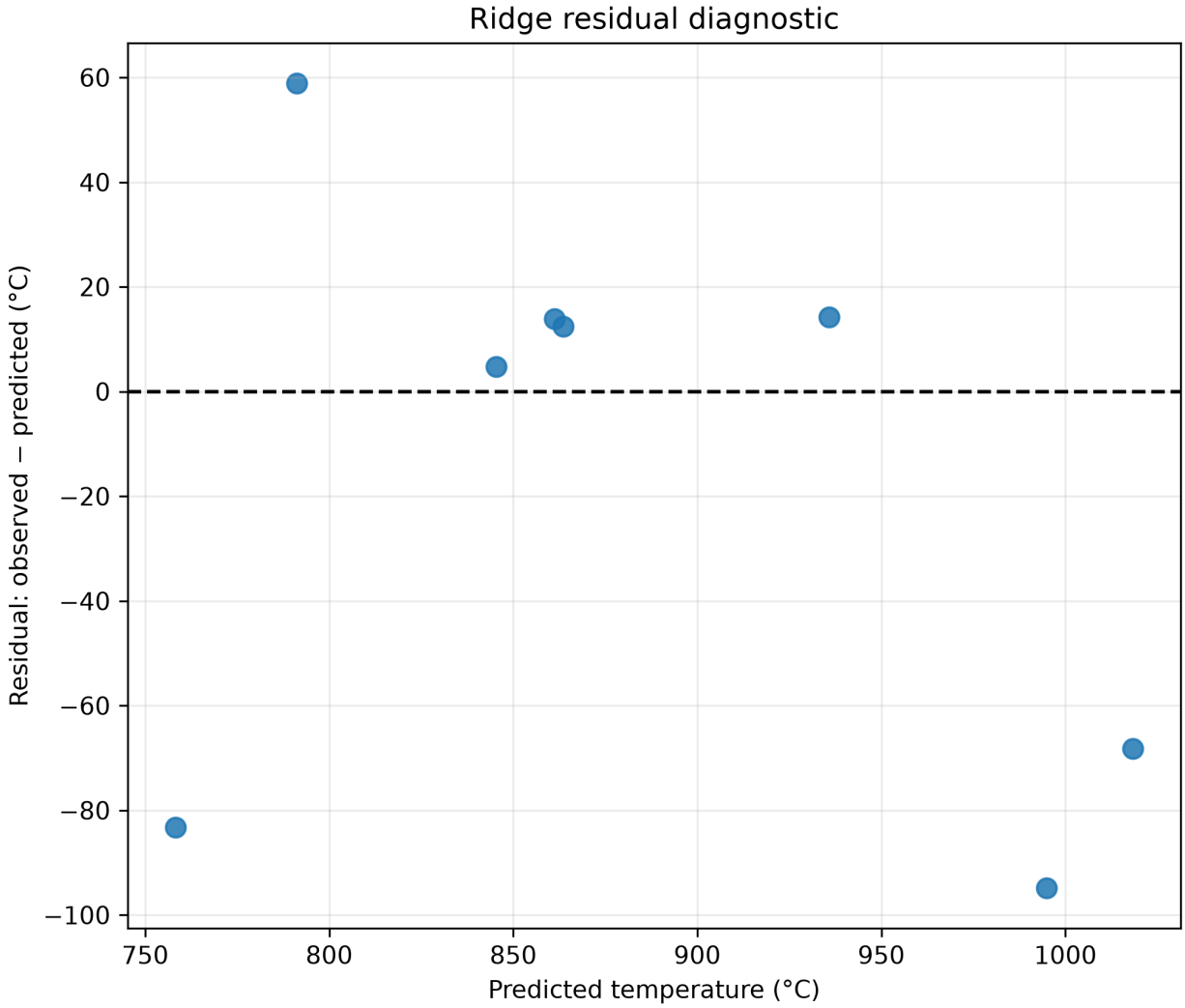


Figure 1. Residuals versus fitted values. Inspect for systematic patterns and changing variance.

The residuals represent observed minus predicted temperatures. Positive residuals indicate underprediction, whereas negative residuals indicate overprediction. The residual pattern suggests that some high-temperature observations are overestimated by the Ridge model. However, given the small validation sample, this pattern should be treated cautiously rather than as conclusive evidence of systematic prediction bias.

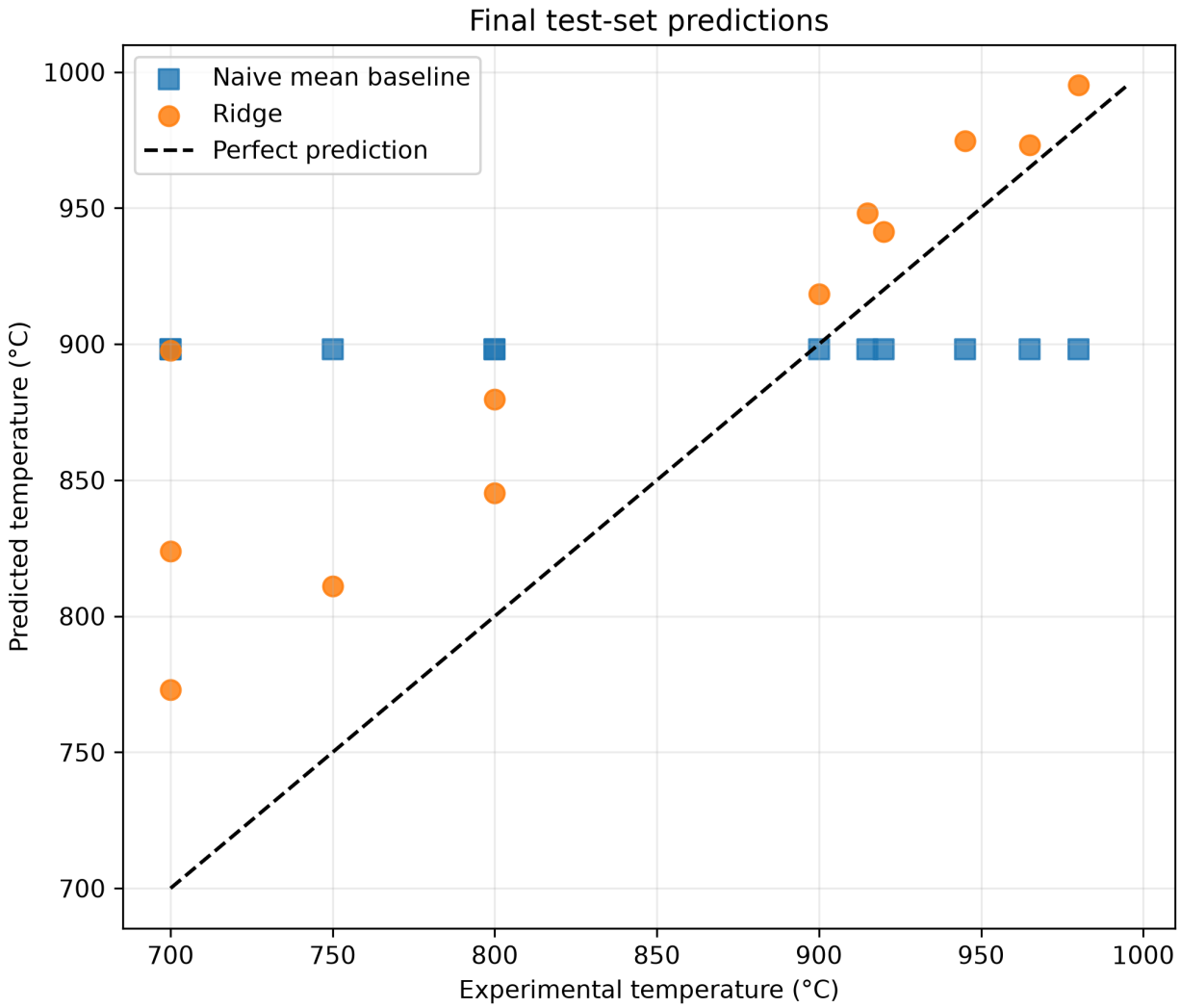


Figure 2. Final held-out test predictions and baseline comparison. Test results were not used to select the model.

The final held-out test results show that the Ridge model outperforms the naive mean baseline in reconstructing experimental temperatures. Ridge achieves a mean absolute error of 58.89 °C and a root mean squared error of 79.04 °C, compared with 97.92 °C and 120.13 °C, respectively, for the baseline. This improvement indicates that mineral chemistry and experimental pressure provide useful predictive information beyond a constant mean temperature. Nevertheless, the test set contains only 12 observations from four experimental studies, and errors remain substantial. The results therefore support the value of the regression approach for this dataset, but do not establish reliable generalisation to all experimental conditions or natural geological samples.

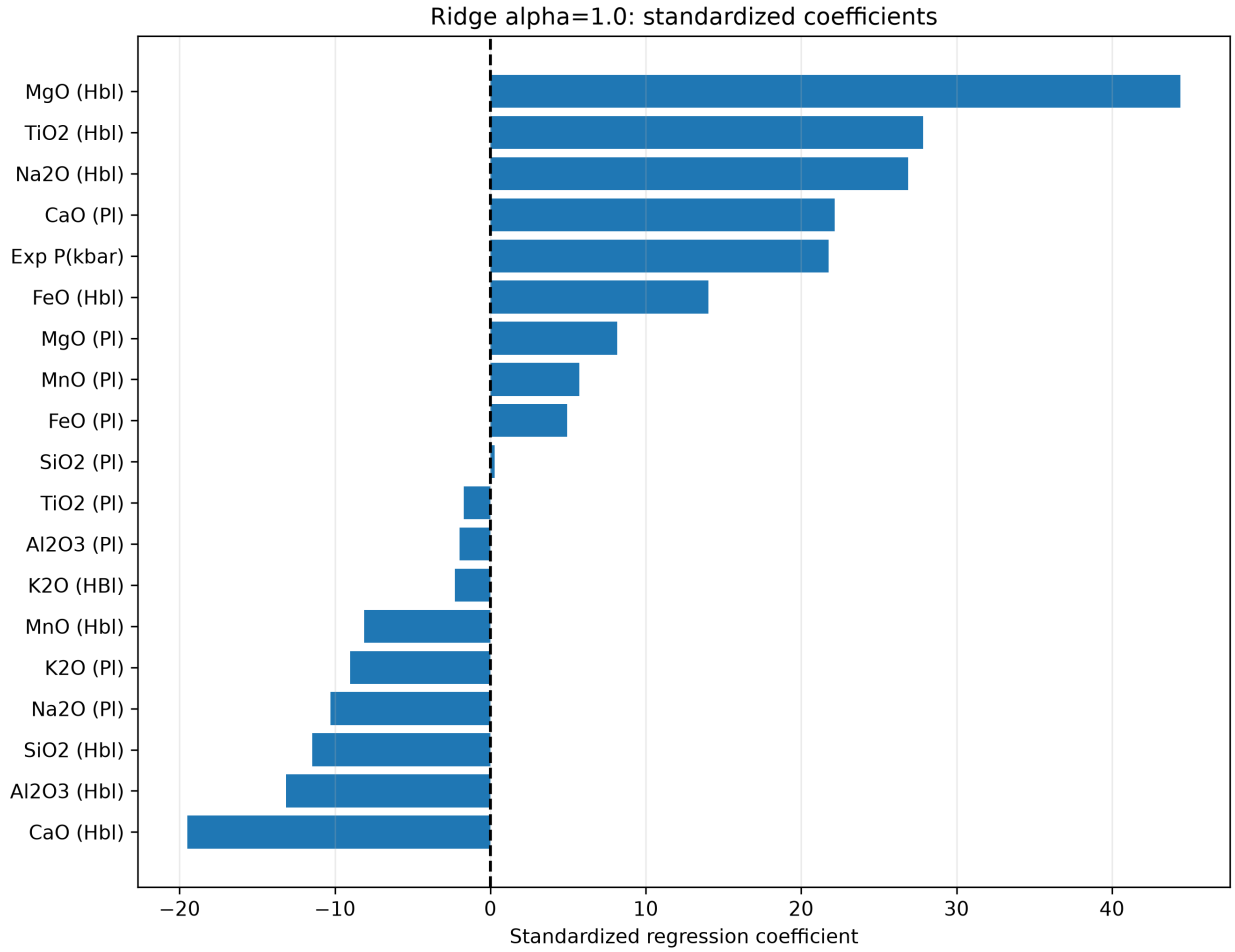


Figure 3. Standardized Ridge coefficients. Coefficients represent conditional associations, not causal effects.

The standardized Ridge coefficients indicate which predictors have stronger positive or negative associations with predicted experimental temperature within the fitted model. Mineral-chemistry variables, including MgO in hornblende and CaO in plagioclase, have relatively large coefficient magnitudes, while experimental pressure also contributes to the prediction. Ridge regularization reduces coefficient magnitudes to help control model complexity when predictors are correlated. However, these coefficients should not be interpreted as independent causal effects, because mineral compositions may be interrelated and the analysis uses a relatively small dataset.

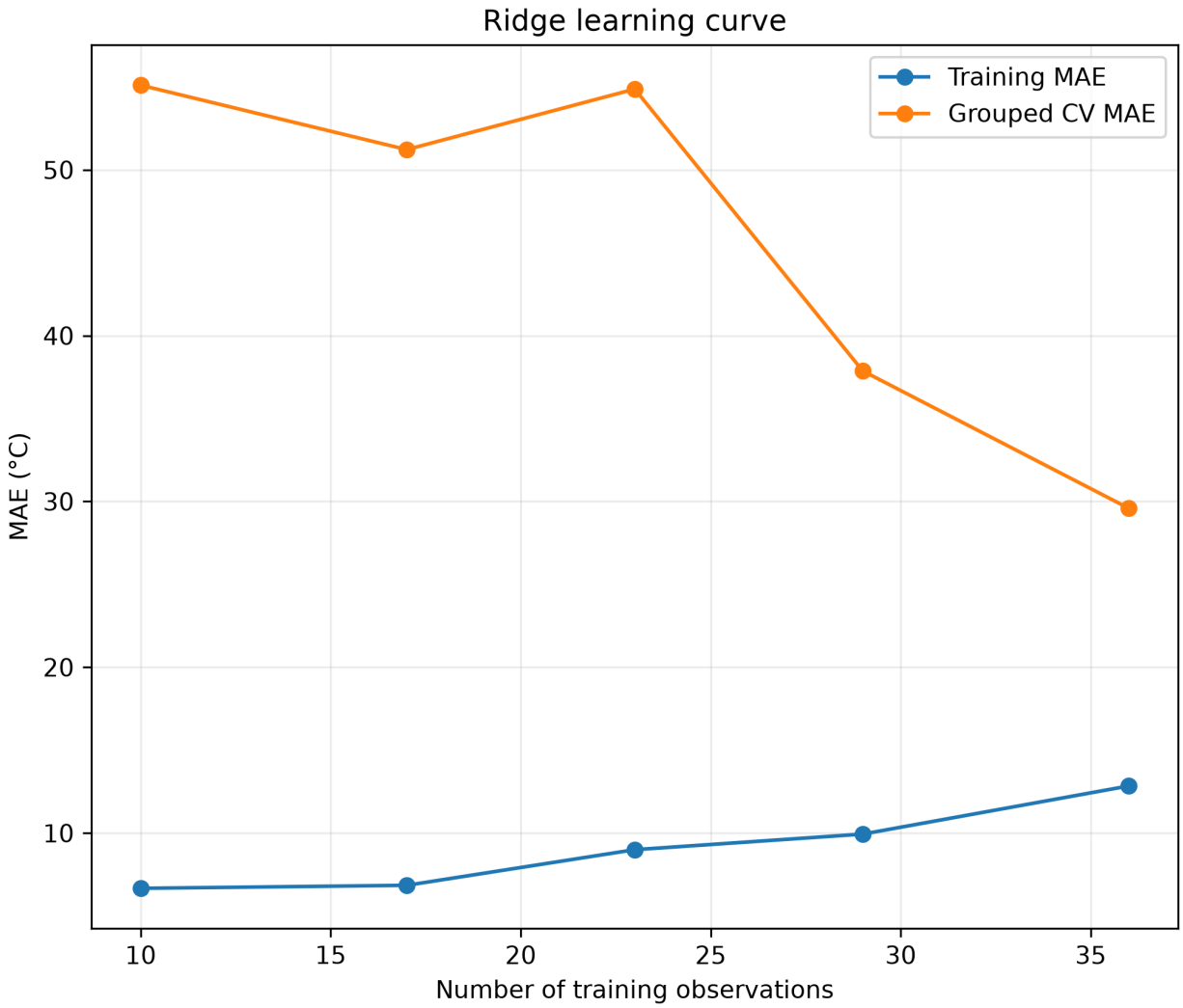


Figure 4. Learning curve. Compare training and cross-validation performance as the training sample size increases. Interpret cautiously because the dataset is small.

The learning curve illustrates how the Ridge model's training and cross-validation errors change as more observations are used for model fitting. A narrowing gap between the two curves may indicate improved generalisation, whereas persistently high validation error suggests that predictive performance remains limited. Because the dataset contains relatively few complete observations distributed across experimental studies, the curve may be sensitive to the particular samples included at each training size. The results should therefore be interpreted as an indication of learning behaviour rather than a definitive estimate of performance on a much larger dataset.

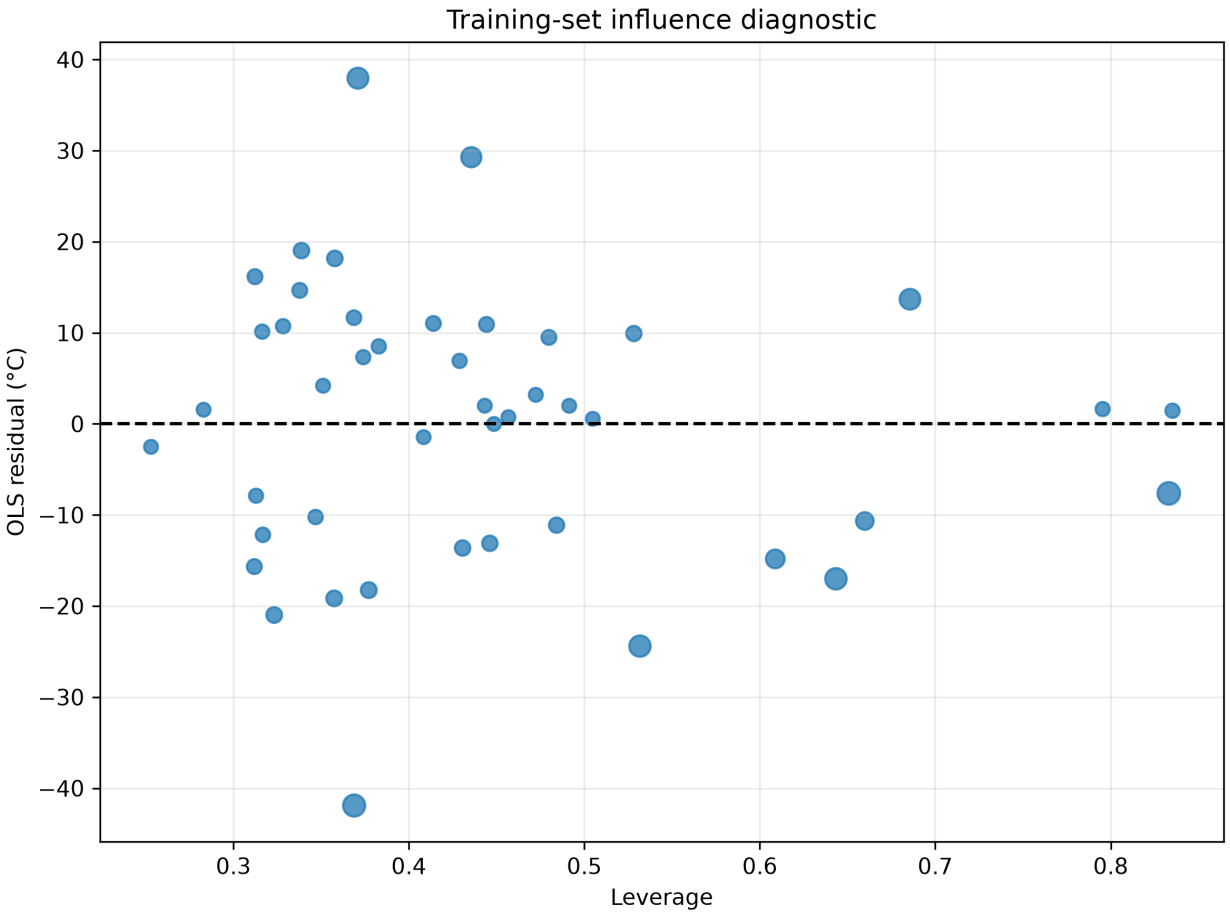


Figure 5. Influence diagnostics for identifying observations that may disproportionately affect the fitted regression.

The influence diagnostics identify observations that may have a disproportionate effect on the fitted regression. Observations with high leverage or large residuals deserve closer examination because they can affect coefficient estimates and predictive performance. However, an influential observation is not necessarily an erroneous measurement; it may represent a genuine experimental condition that is underrepresented in the dataset. These cases should be checked against their source records before any decision to exclude them. Their presence also highlights the need for caution when interpreting a model fitted to a small number of experimental runs.

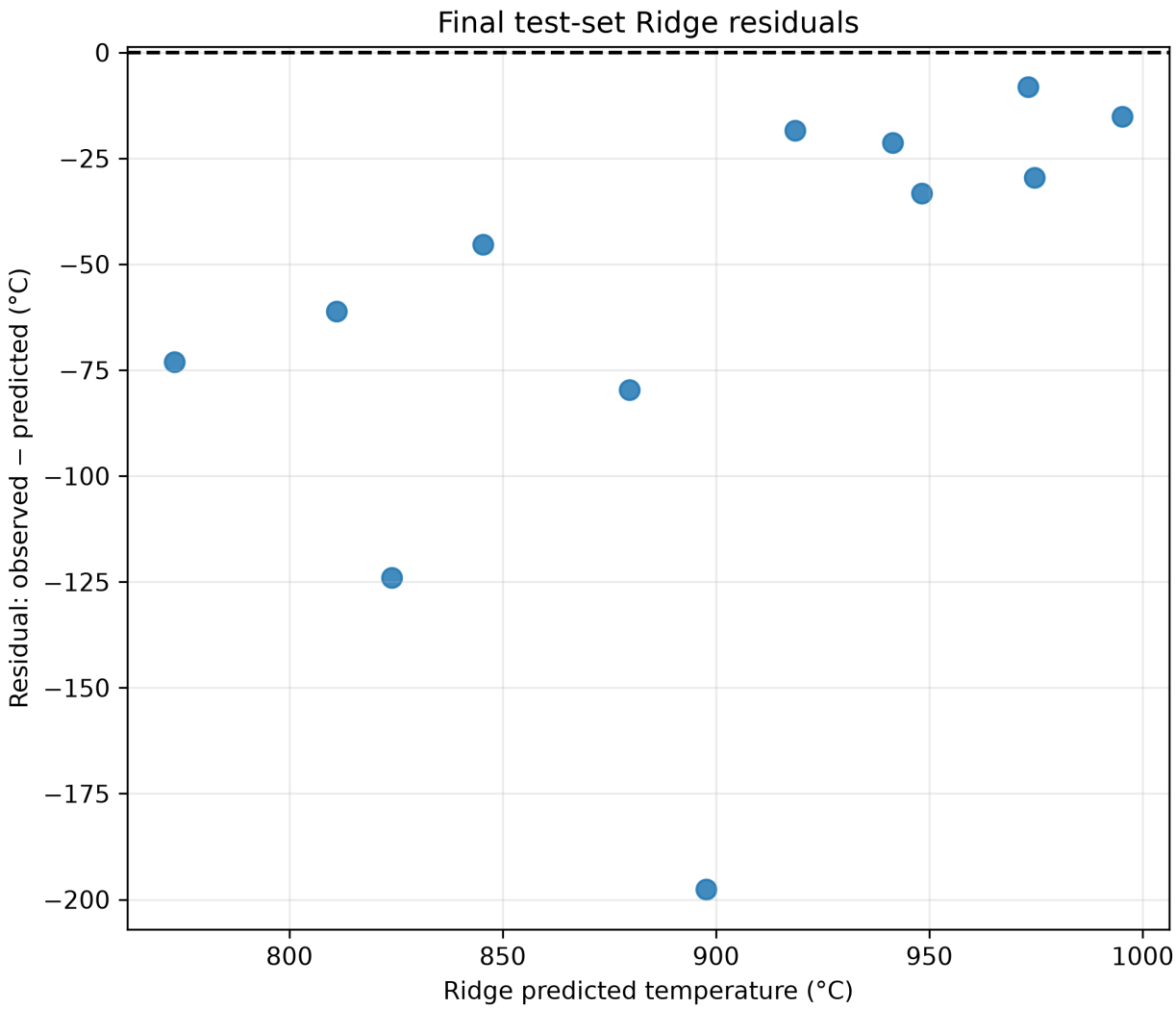


Figure 6. Final held-out test residuals. Residuals are observed minus predicted temperatures; the test set is used for evaluation, not model selection.

The test residuals are calculated as observed minus predicted experimental temperature. Positive residuals indicate underprediction, whereas negative residuals indicate overprediction. The spread of residuals shows that the Ridge model does not predict every test observation with the same accuracy. Some errors remain substantial, so the improvement over the naive baseline should not be interpreted as precise reconstruction for every experimental run. Because the test set contains only 12 observations from four studies, these results provide a limited assessment of performance on entirely new experimental sources.

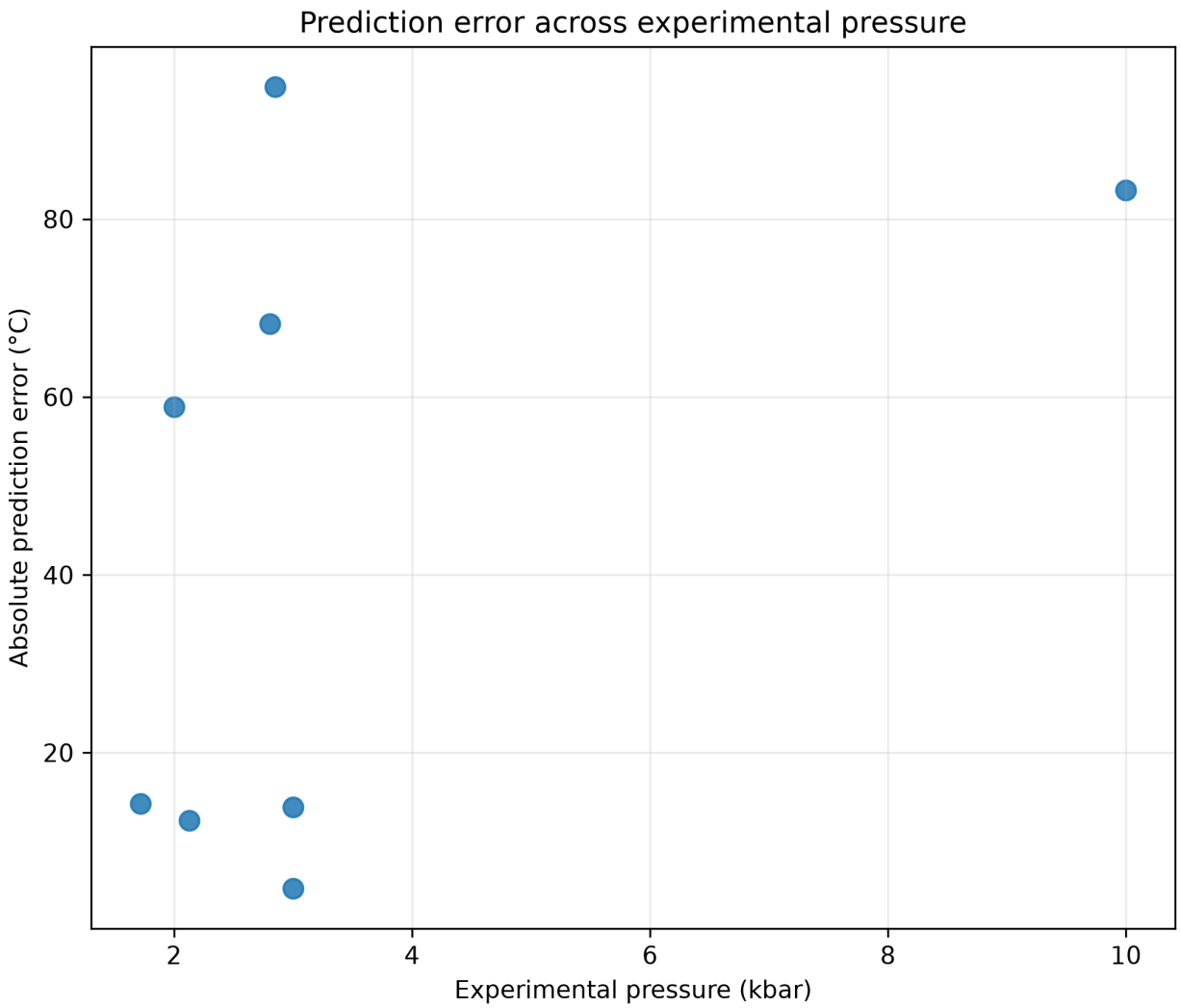


Figure 7. Regression prediction error versus experimental pressure.

Heteroscedasticity assessment. The Breusch-Pagan test returned an LM statistic of 2.2716 and p = 0.1318. At the conventional 0.05 significance level, this test does not provide sufficient evidence to reject the null hypothesis of constant error variance. This is not proof that the variance is constant, particularly given the small dataset.

Final held-out test metrics

model	MAE_C	RMSE_C	R2
Naive mean baseline	97.917	120.125	-0.312
Ridge alpha=1.0	58.894	79.043	0.432

Held-out test errors by study

study_id	n_runs	ridge_MAE_C	ridge_bias_C
apted_1983	1	197.618	-197.618
ernst_liu_1998	6	66.943	-66.943
grove_1997	4	18.565	-18.565
krawczynski_2012	1	33.194	-33.194

3. PH94 and PH21 Error Analysis

The thermometer comparison includes 65 matched observations. The reported overall MAE was 58.43 °C for PH94 and 61.40 °C for PH21. Mean signed error was 3.20 °C for PH94 and 52.52 °C for PH21. These summaries describe this dataset and do not establish that either thermometer will perform similarly in other geological settings.

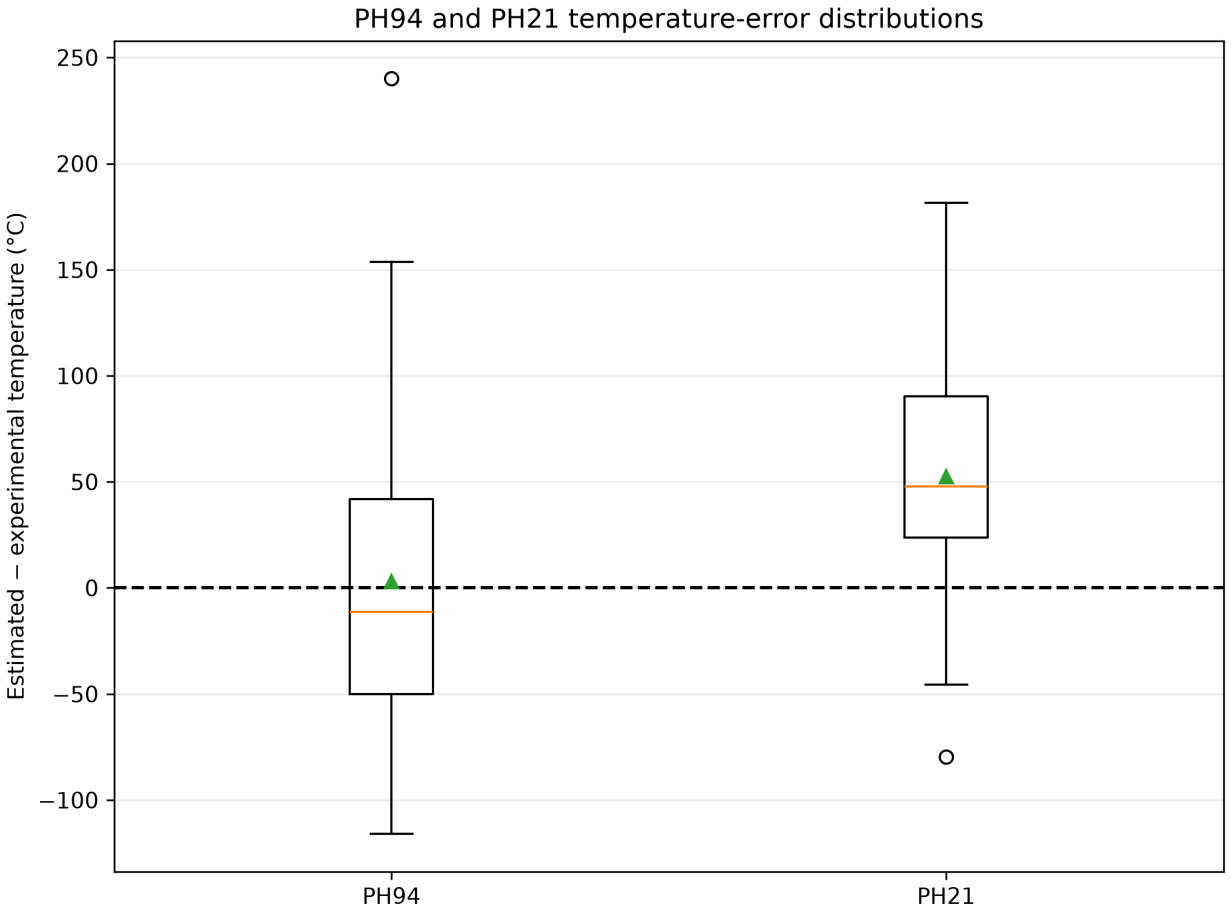


Figure 8. Distributions of PH94 and PH21 errors

The error distributions compare the temperature estimates from PH94 and PH21 against experimental reference temperatures. PH94 has a mean absolute error of 58.43 °C, compared with 61.40 °C for PH21, indicating a slightly smaller average error magnitude for PH94 in these matched observations. Their mean signed errors also differ: 3.20 °C for PH94 and 52.52 °C for PH21, suggesting a stronger overall tendency toward overestimation for PH21. These aggregate differences do not imply that one method performs better for every experimental source or condition.

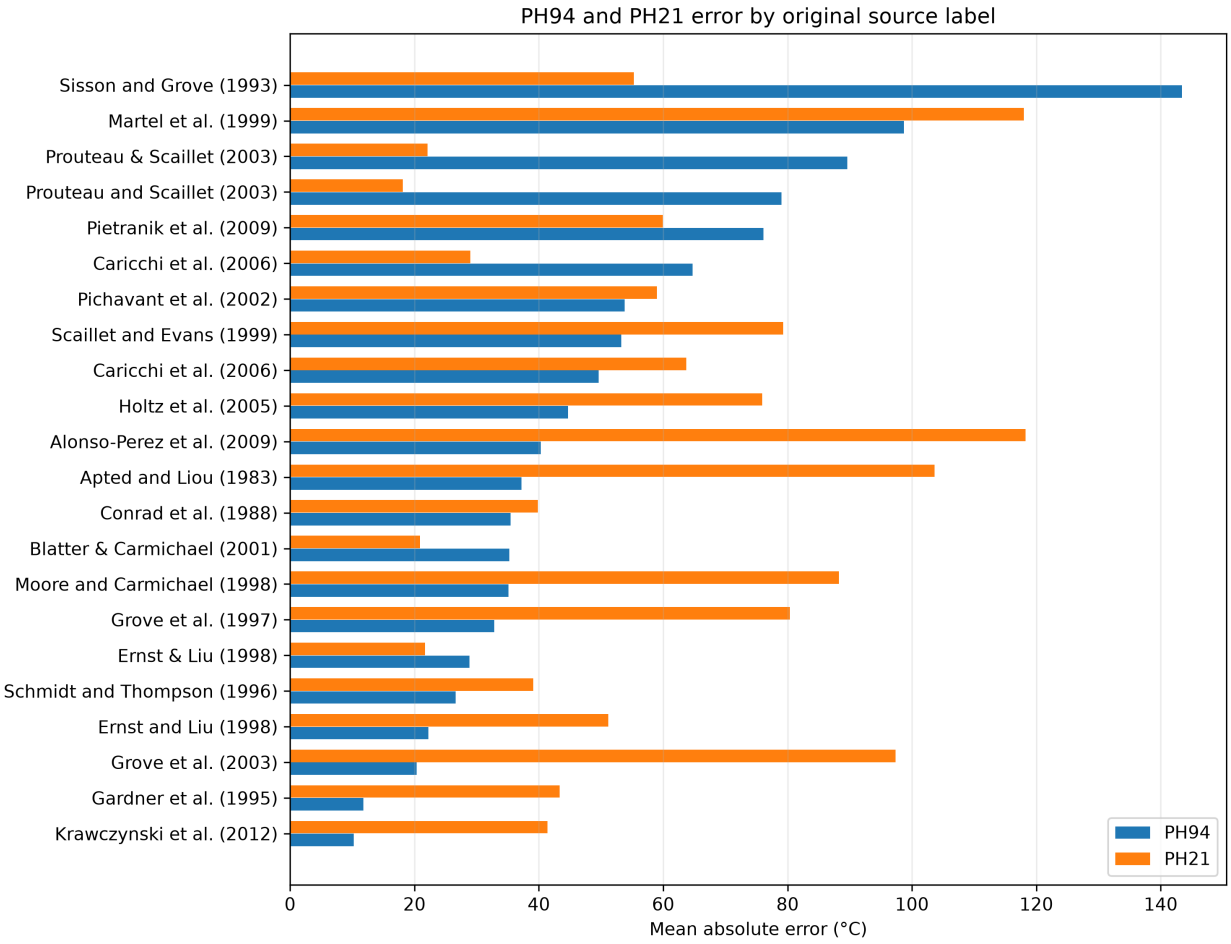


Figure 9. Source-level mean absolute errors. Some bibliographic aliases still require reconciliation.

The source-level comparison shows that thermometer error magnitude varies across the experimental sources represented in the dataset. This suggests that the overall performance metrics can conceal important differences between groups of experiments. However, the number of runs per source varies considerably, and some sources contribute only one observation. In addition, alternative bibliographic labels may refer to the same publication. Source-specific comparisons should therefore remain descriptive until the source mapping is fully reconciled and the small group sizes are considered.

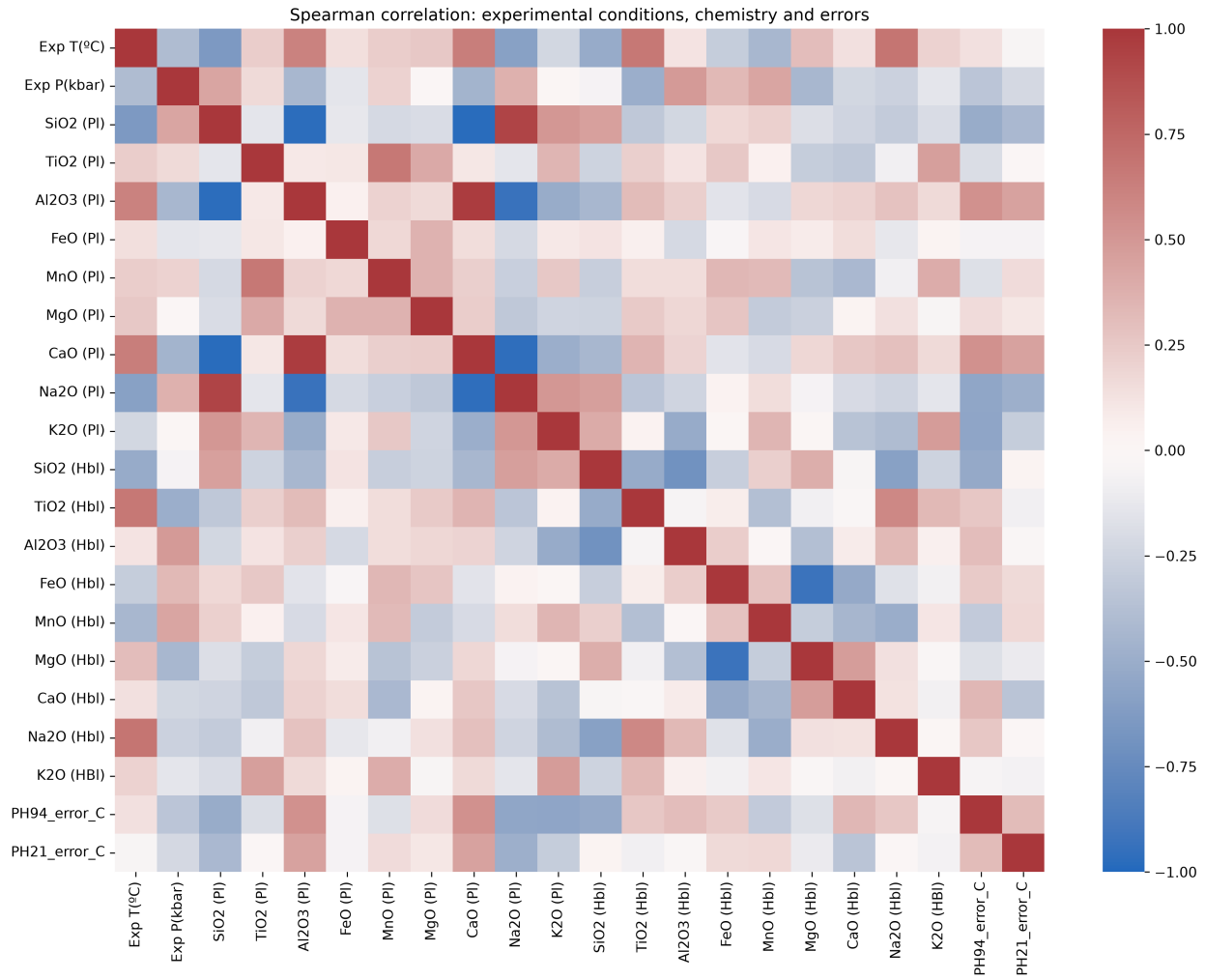


Figure 10. Exploratory associations between selected variables and thermometer errors.

The correlation analysis explores associations between mineral chemistry and the signed or absolute errors of the two geothermometers. Several plagioclase composition variables show associations with signed errors, suggesting that error direction may vary with mineral composition. However, signed error and absolute error describe different properties: the former captures bias, while the latter measures error magnitude. Mineral-chemistry variables may also be correlated with one another, and observations from the same study may not be independent. These exploratory associations do not establish causal relationships.

Overall thermometer metrics

method	n	mean_signed_error_C	median_signed_error_C	MAE_C	RMSE_C
PH94	65	3.196	-11.278	58.430	72.820
PH21	65	52.516	47.880	61.404	72.498

Thermometer errors by source

source_original	n	PH94_bias_C	PH94_MAE_C	PH21_bias_C	PH21_MAE_C
Sisson and Grove (1993)	7	143.394	143.394	52.718	55.276
Scaillet and Evans (1999)	7	-53.289	53.289	79.267	79.267
Pichavant et al. (2002)	6	29.817	53.770	59.004	59.004
Prouteau and Scaillet (2003)	6	-79.028	79.028	1.918	18.103
Alonso-Perez et al. (2009)	5	17.878	40.345	118.227	118.227
Grove et al. (1997)	4	32.828	32.828	80.375	80.375
Pietranik et al. (2009)	3	76.101	76.101	53.761	59.961
Moore and Carmichael (1998)	3	-17.418	35.135	67.914	88.250
Ernst, W.G., Liu, J., 1998. Experimental phase-equilibr	3	2.901	28.825	21.675	21.675
Holtz et al. (2005)	3	-44.707	44.707	75.896	75.896
Caricchi et al. (2006)	3	-64.704	64.704	-28.970	28.970
Ernst and Liu (1998)	3	22.243	22.243	49.191	51.180

Exploratory error-feature associations

feature	error_metric	n	spearman_rho	p_value_unadjusted	p_value_BH
Exp T(ºC)	PH94_error_C	65	0.136	0.282	0.424
Exp T(ºC)	PH94_abs_error_C	65	0.190	0.130	0.266
Exp T(ºC)	PH21_error_C	65	-0.045	0.722	0.850
Exp T(ºC)	PH21_abs_error_C	65	0.099	0.435	0.579
Exp P(kbar)	PH94_error_C	65	-0.337	0.006	0.024
Exp P(kbar)	PH94_abs_error_C	65	-0.190	0.129	0.266
Exp P(kbar)	PH21_error_C	65	-0.226	0.071	0.172
Exp P(kbar)	PH21_abs_error_C	65	-0.225	0.071	0.172
SiO2 (Pl)	PH94_error_C	65	-0.515	0.000	0.000
SiO2 (Pl)	PH94_abs_error_C	65	-0.247	0.047	0.126
SiO2 (Pl)	PH21_error_C	65	-0.423	0.000	0.003
SiO2 (Pl)	PH21_abs_error_C	65	-0.430	0.000	0.002

4. Interpretation and Limitations

On validation data, Ridge achieved an MAE of 43.81 °C and RMSE of 55.48 °C, compared with a naive-baseline MAE of 61.22 °C and RMSE of 89.52 °C.

On the held-out test set of 12 runs from four studies, Ridge achieved an MAE of 58.89 °C, RMSE of 79.04 °C, and R^2 of 0.432. The naive baseline achieved an MAE of 97.92 °C, RMSE of 120.13 °C, and R^2 of -0.312. The small number of independent test studies limits the precision and generalizability of this comparison.

PH94 showed a smaller overall MAE than PH21 in the matched sample, while PH21 had a larger positive mean signed error. Error patterns differed across source labels. These findings are descriptive; the source mapping must be verified before making strong study-specific claims.

Important limitations include the small grouped test set, complete-case selection, correlated mineral chemistry predictors, uncertain source aliases, and the possibility that experimental conditions differ from natural geological settings. Correlation and regression coefficients do not prove causal effects.

5. Reproduction and Transparency

The project includes numbered Python scripts in src/, processed data in data_processed/, result tables in results/, and diagnostic figures in figures/. The environment audit records Python and package versions in results/requirements-lock.txt and results/environment-details.txt.

Before submission, verify that the original supplementary dataset and its version are documented, reconcile source aliases, and confirm that the scripts run from a clean environment. The environment audit records the current environment and does not alone prove bit-for-bit reproducibility.

AI assistance supported code scaffolding, debugging guidance, methodological discussion, and documentation. The accompanying AI_USE_LOG.md records the intended disclosure and verification responsibilities. The student is responsible for checking all data provenance, outputs, interpretations, and final claims.